HOSPITAL PATIENT ANALYTICS

Executive Overview & Data Insights

JUNE 2026 • OPERATIONS • COSTS • OUTCOMES

40,596 cleaned records | 21 fields | 8 dashboard visuals

Prepared 16 October 2026 | Analytics team

Source workbook and project data files

Dataset scope & measures

One month of descriptive patient-record analysis

SCOPE

40,596

records after cleaning

- 21 source fields
- Coverage: 1–30 June 2026
- 40,000 distinct patient IDs
- No unique encounter ID

CORE FIELDS

- Demographic and patient identifiers
- Disease, severity, referral, status
- Service, dates, costs, payment
- Wait, room time, insurance, rating

EXECUTIVE METRICS

30,759

OPD records

9,837

IPD records

? 1.10B

total recorded treatment cost

3.96 / 5

average patient rating

Counting caveat — **Records are not confirmed unique visits; repeated IDs lack an encounter key.**

Data cleaning & quality checks

Raw extract: 40,600 rows → cleaned extract: 40,596 rows

PROFILE

4

exact duplicates removed

- 85 ages outside 0–120
- No raw missing values
- 21 fields; 40,600 source rows

TRANSFORM

85

invalid ages imputed; mean 43

- Parsed dates and numeric fields
- Normalized text; kept other values

VALIDATE

0

missing values remain

- Date order checks passed
- OPD/IPD timing checks passed
- 40,596 output rows retained

WHY MEAN

Near-symmetric ages (skew −0.107; mean 43.33; median 44) suited mean imputation. Median suits skewed numeric data; mode fills categorical gaps.

Data limitation

1,192 rows share patient IDs; repeated IDs may be separate encounters.

Additional preprocessing & validation

Operations performed, including checks that required no correction

NORMALIZE & TYPE

Text, numeric fields, dates

- Trimmed text and normalized blanks; 0 values changed.
- Converted five numeric fields; parsed both date fields.
- Exported dates in ISO format.
- No categorical gaps; mode imputation unnecessary.

RANGE CHECKS

Completed on cleaned records

- Checked positive costs and ratings within 1–5.
- Checked nonnegative service times and valid date order.
- No post-cleaning range or date violations remained.

RELATIONSHIPS & PRESERVATION

Keep source meaning intact

- OPD/IPD timing rules: 0 mismatches.
- Emergency category and referral checks: 0 mismatches.
- Retained repeated IDs with differing records, age zero, and high costs.

WHY KEEP THESE CHECKS?

Validation documents what was tested; passed checks required no edits.

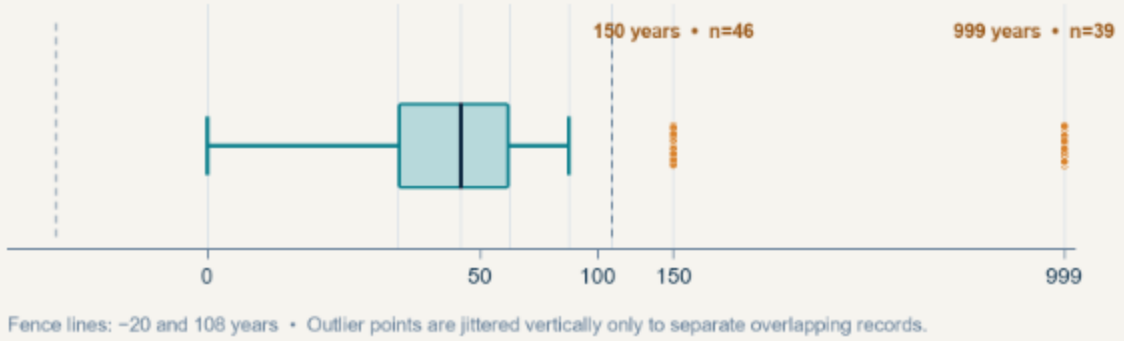
Age outliers in the uncleaned dataset

Box plot only | clean_patients.csv | 40,600 age values

UNCLEANED AGE DISTRIBUTION

40,600 recorded ages | IQR outliers highlighted

Horizontal box plot; compressed scale above typical ages keeps 999 visible



Box spans the middle 50%; whiskers stop at in-fence values. The age axis is compressed above typical values.

IQR CALCULATION

Q1 = 28 • Median = 44 • Q3 = 60

IQR = Q3 − Q1 = 60 − 28 = 32 years

Fences: 28 − 1.5×32 = −20 | 60 + 1.5×32 = 108

SIX BOXPLOT ELEMENTS

Lower whisker	**0 years**
Q1	**28 years**
Median	**44 years**
Q3	**60 years**
Upper whisker	**85 years**
Outliers	**85 rows: 150×46; 999×39**

INTERPRETATION

85/40,600 = 0.21%; all 85 exceed 108, none fall below −20. Review before correction.

Daily patient records

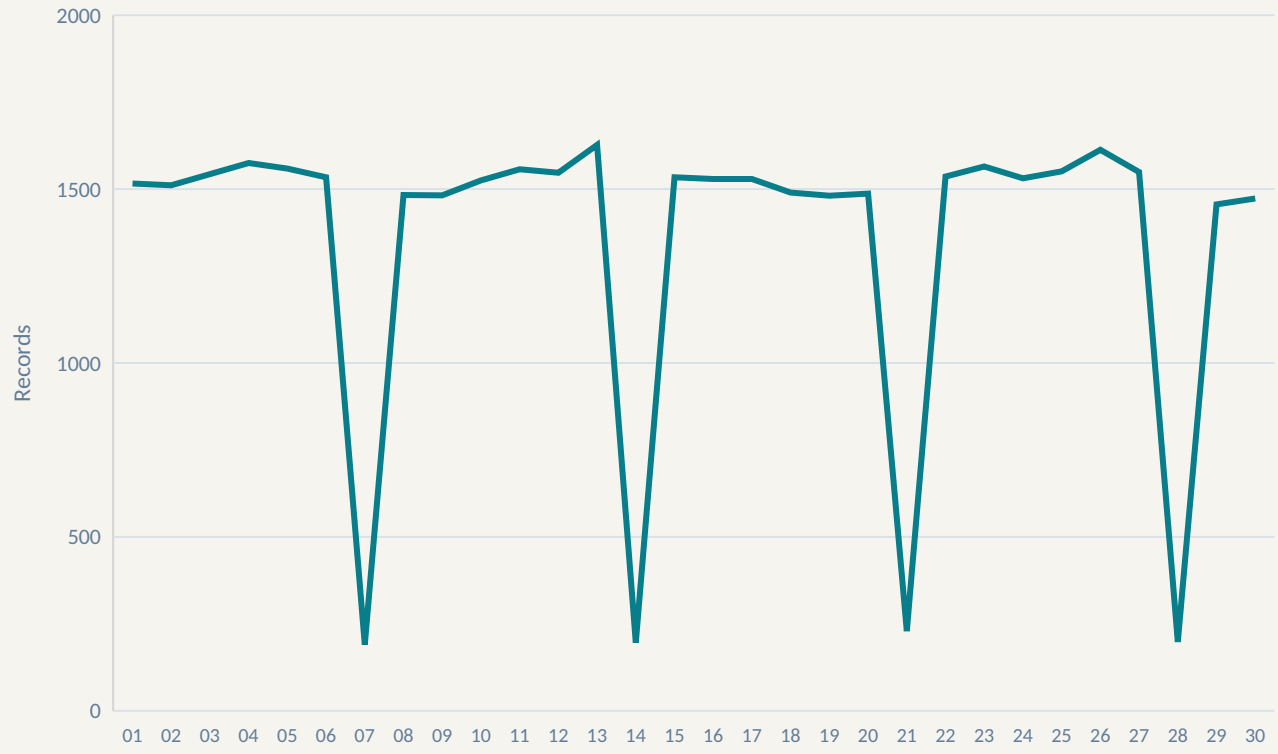
Line chart | Admissions by date, June 2026

Primary insight

Most dates cluster near 1,500 records; four Sundays fall sharply.

KEY OBSERVATIONS

- Median: 1,529 records per day.
- Peak: 1,627 records on 13 June.
- Sundays: 190–229 records, about 85% below median.



DECISION IMPLICATION

Verify Sunday reporting and operating schedules before changing staffing.

OPD and IPD service mix

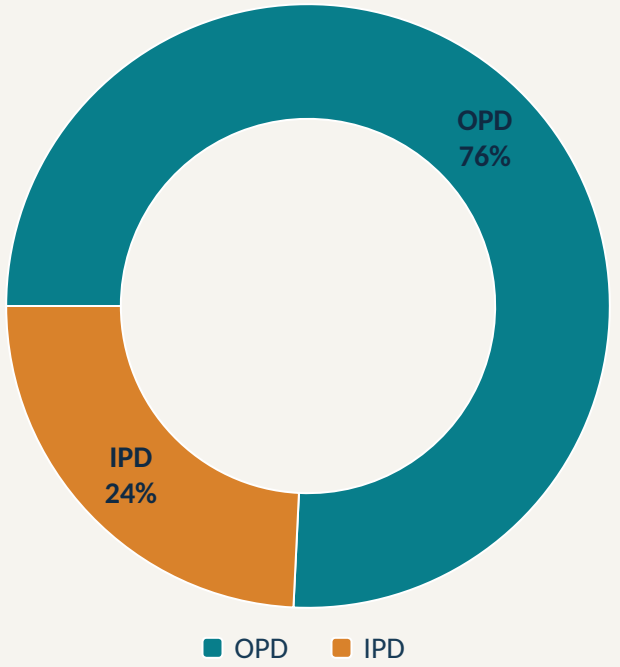
Doughnut chart | Share of cleaned records

Primary insight

OPD represents about three-quarters of recorded service volume.

KEY OBSERVATIONS

- OPD: 30,759 records (75.8%).
- IPD: 9,837 records (24.2%).
- Record mix does not establish capacity utilization.



DECISION IMPLICATION

Align outpatient capacity to volume; protect inpatient staffing and beds.

Recorded patient outcomes

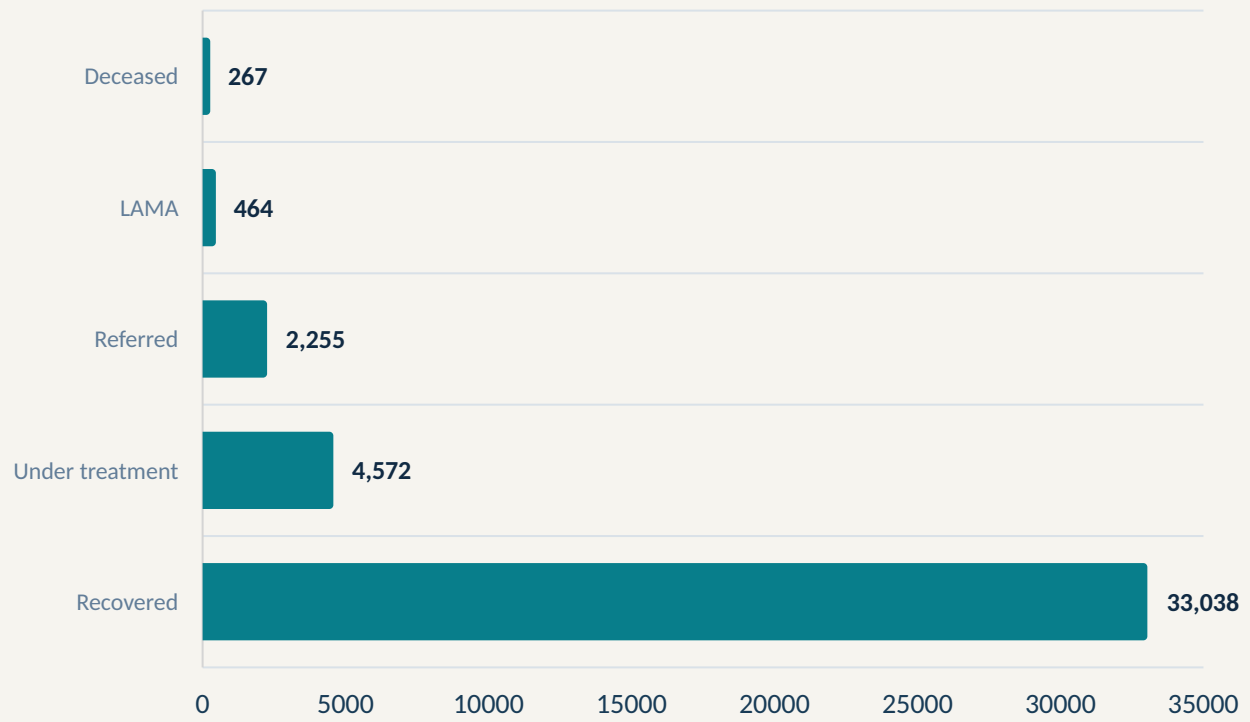
Horizontal bar chart | Status distribution

Primary insight

Recovery is the most common recorded status; 18.6% show another status.

KEY OBSERVATIONS

- Recovered: 33,038 records (81.4%).
- Under treatment: 4,572; referred: 2,255.
- LAMA: 464; deceased: 267 (0.7%).



DECISION IMPLICATION

Review non-recovered statuses in clinical context; avoid risk-adjusted claims.

Severity distribution

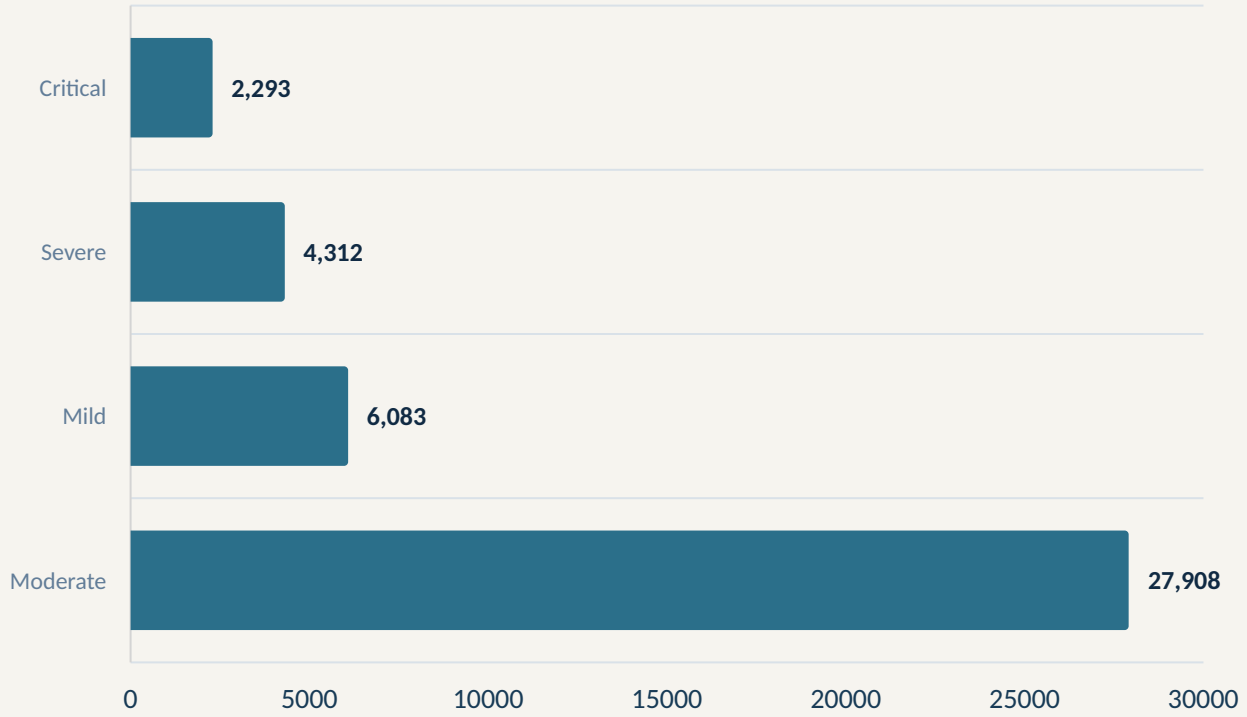
Horizontal bar chart | Recorded case severity

Primary insight

Moderate cases dominate the recorded severity mix.

KEY OBSERVATIONS

- Moderate: 27,908 records (68.7%).
- Mild: 6,083; severe: 4,312.
- Critical: 2,293; severe plus critical: 16.3%.



DECISION IMPLICATION

Use acuity mix in capacity reviews; confirm the source coding definitions.

Average treatment cost per patient-day

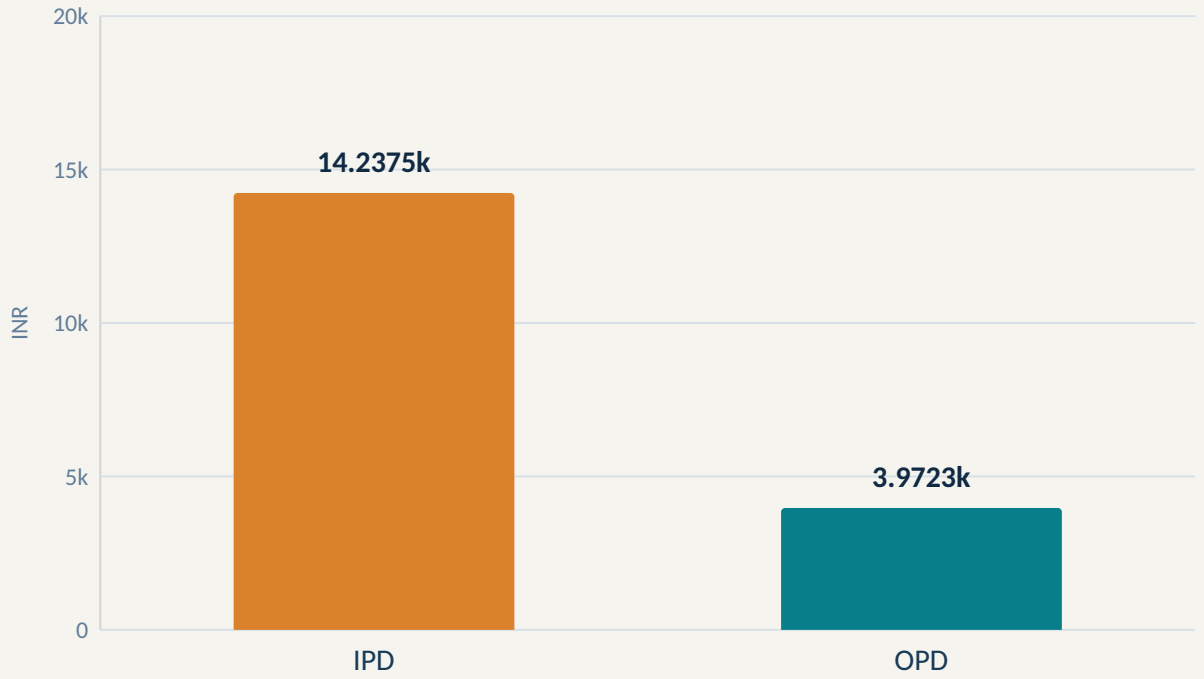
Column chart | OPD and IPD, INR

Primary insight

Recorded IPD cost per patient-day is about 3.6× OPD.

KEY OBSERVATIONS

- IPD: ₹ 14,237.55 per patient-day.
- OPD: ₹ 3,972.30 per patient-day.
- Compare only after confirming cost and day definitions.



DECISION IMPLICATION

Review inpatient cost drivers and validate patient-day calculations.

Average service time

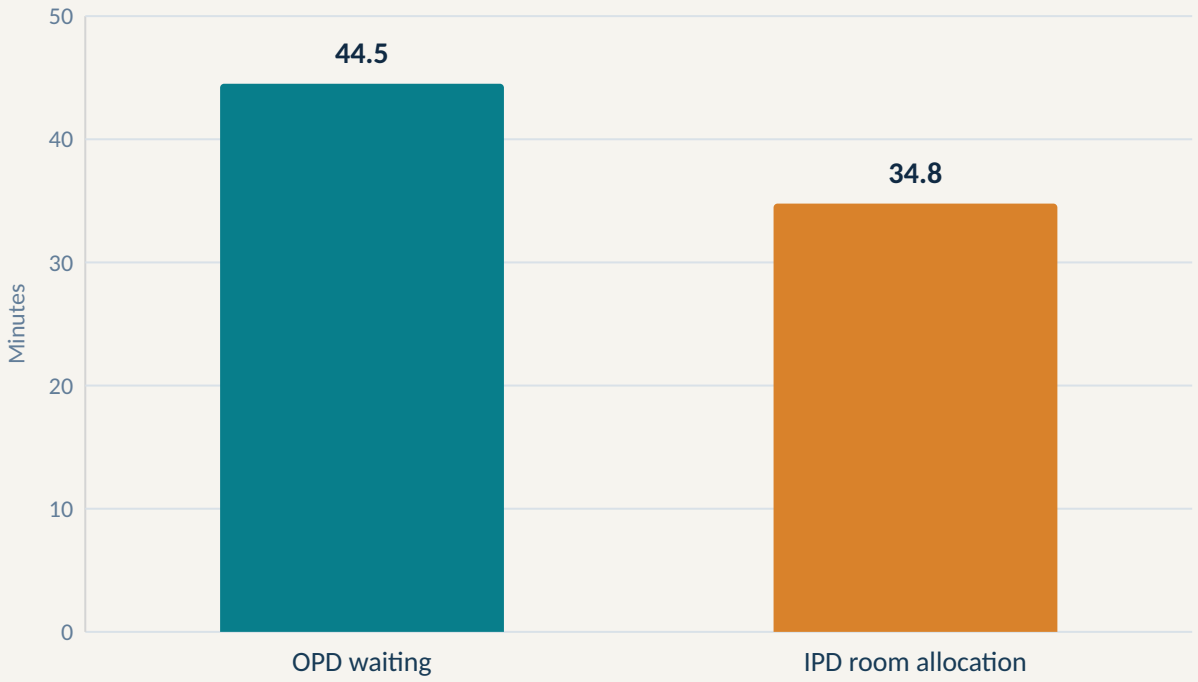
Column chart | Service-specific minutes

Primary insight

OPD waiting and IPD room allocation average distinct service times.

KEY OBSERVATIONS

- OPD average waiting time: 44.5 minutes.
- IPD average room allocation: 34.8 minutes.
- Do not compare as equivalent workflow measures.



DECISION IMPLICATION

Set separate service targets and track monthly median alongside averages.

Average recorded treatment cost by payment method

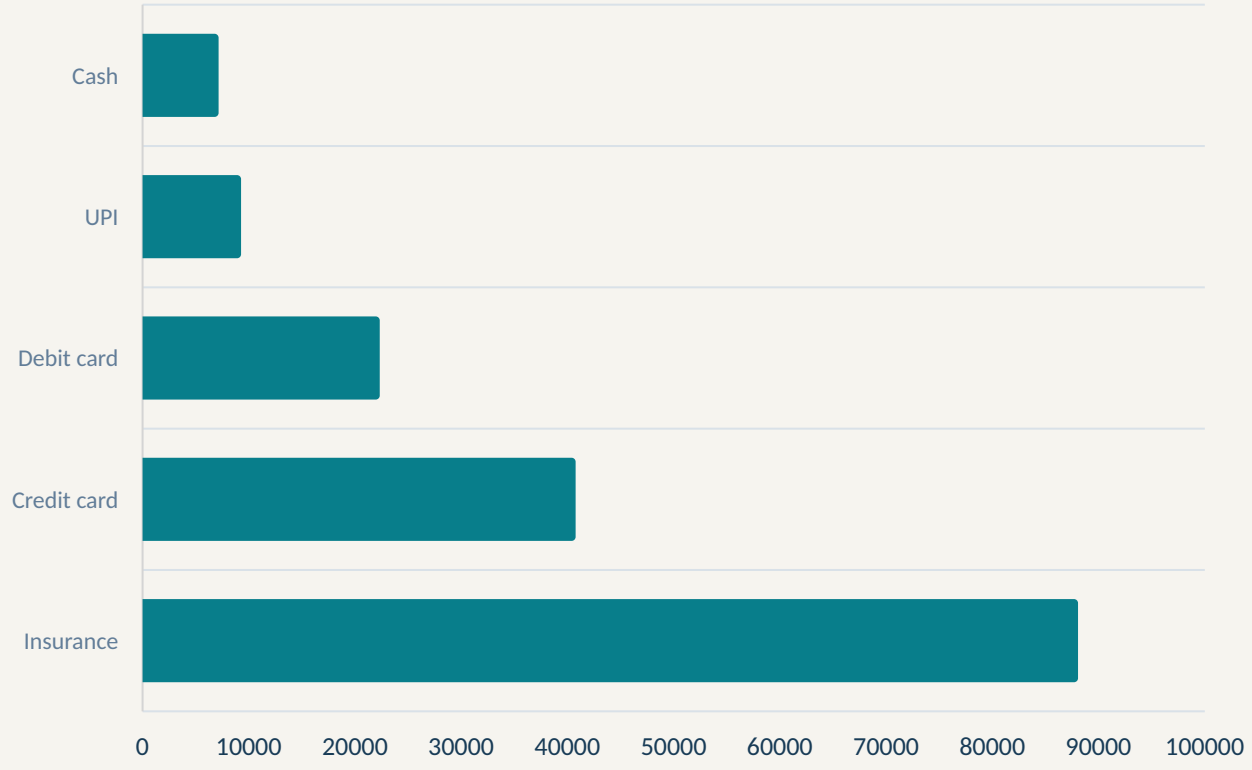
Horizontal bar chart | INR per record

Primary insight

Average recorded cost varies by payment label and case mix.

KEY OBSERVATIONS

- Insurance: ₹ 88,064 average; credit card: ₹ 40,767.
- Debit: ₹ 22,327; UPI: ₹ 9,272; cash: ₹ 7,157.
- Averages may reflect service mix and high-cost cases.



DECISION IMPLICATION

Compare payment categories within service type before drawing conclusions.

Emergency records by insurance status

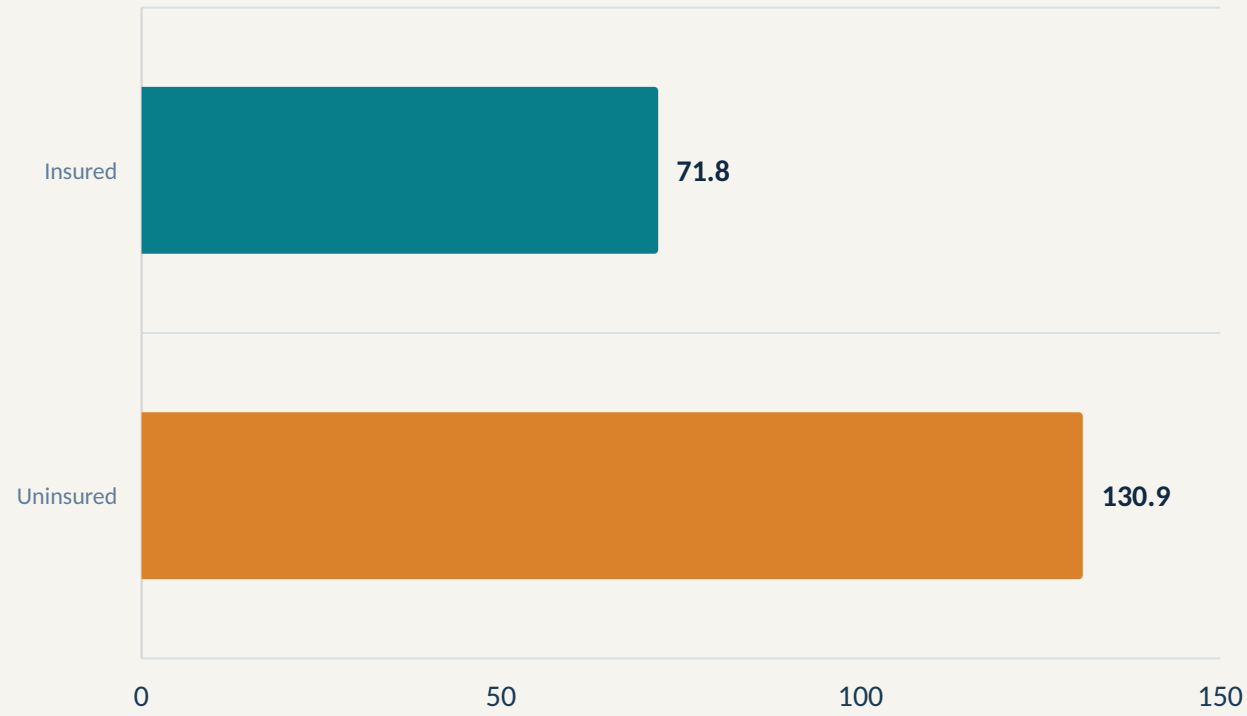
Horizontal bar chart | Average records per day, June 2026

Primary insight

More daily emergency records are marked uninsured in this extract.

KEY OBSERVATIONS

- Uninsured: 130.9 emergency records per day.
- Insured: 71.8 emergency records per day.
- Validate insurance capture and case mix before action.

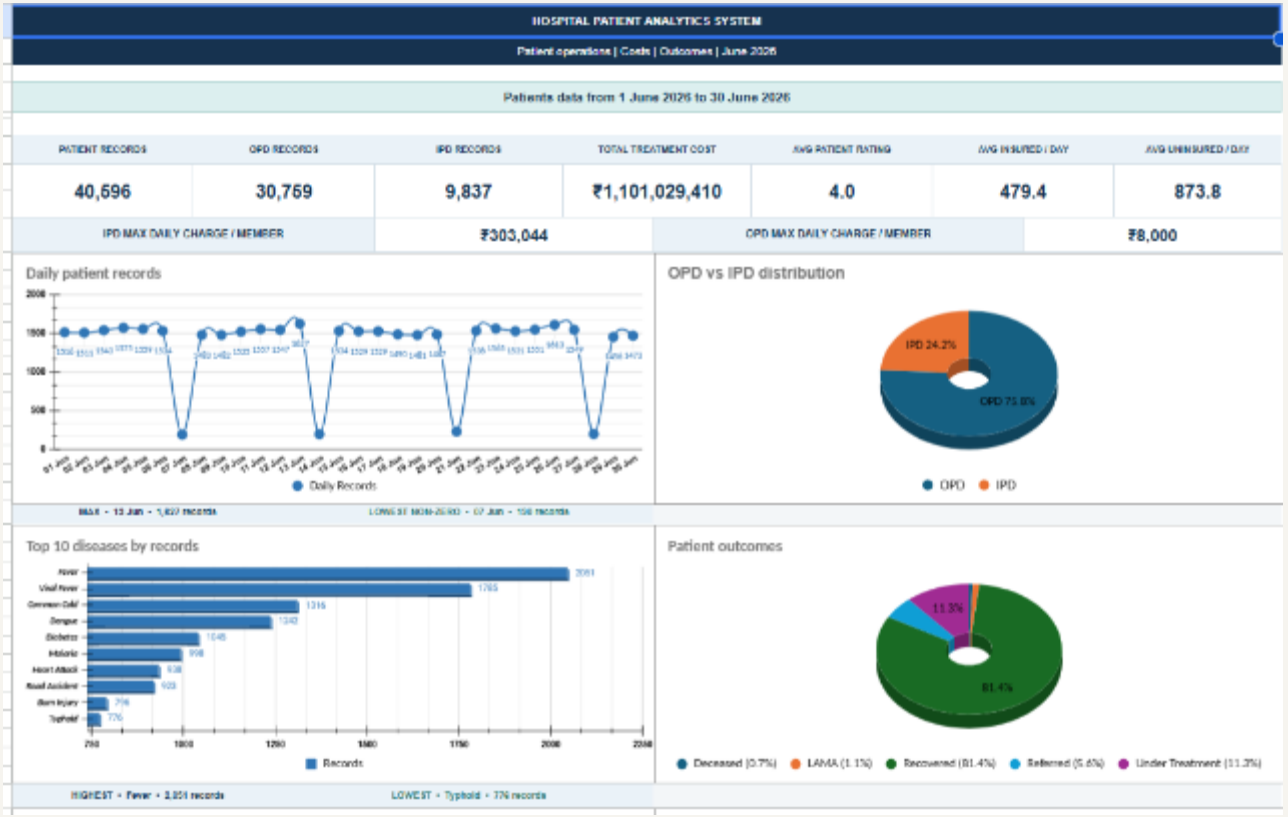


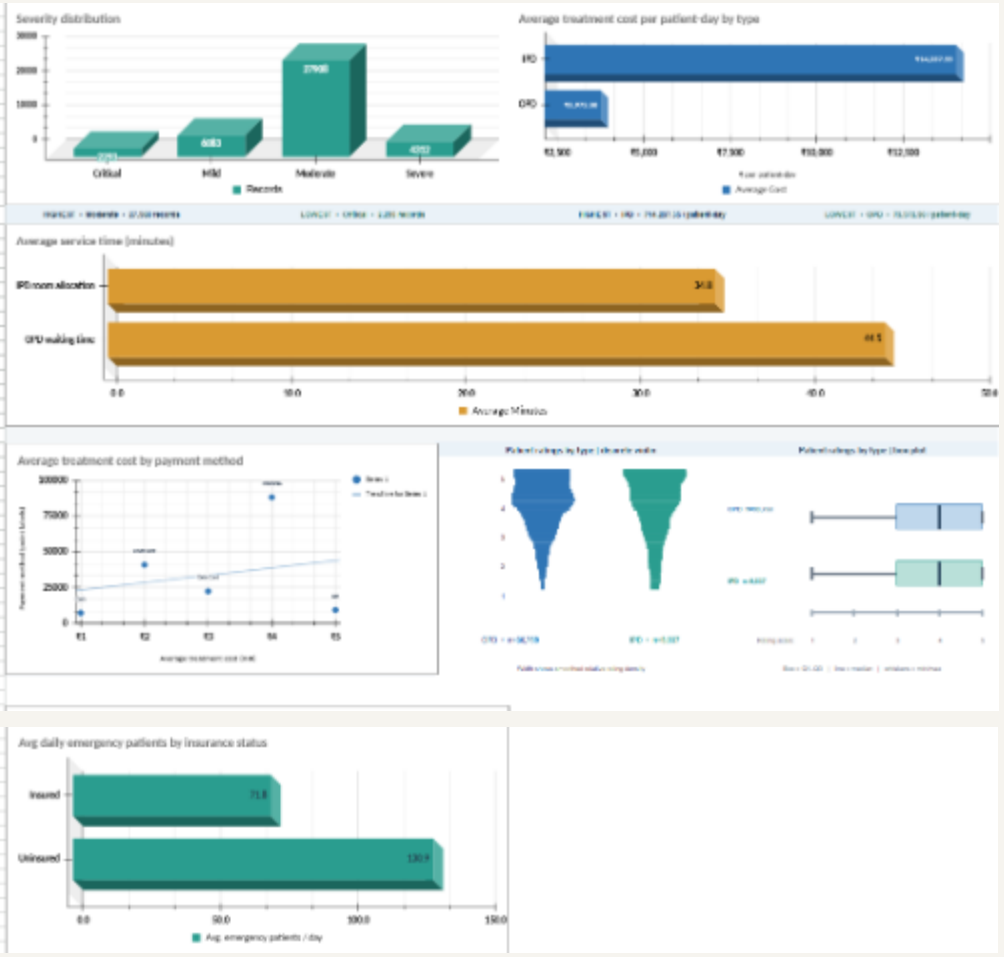
DECISION IMPLICATION

Check insurance completeness and emergency case mix before resource decisions.

Executive dashboard overview

Dashboard screenshots supplied by the user





Conclusion & strategic goal

CORE CONCLUSION

June provides a one-month baseline; validate reporting anomalies before operational changes.

RECOMMENDED ACTIONS

- Verify Sunday volume and insurance capture with source-system owners.
- Segment cost comparisons by service type and clinical severity.
- Set separate targets for OPD waiting and IPD room allocation.

DELIVERED

40,596 cleaned records • 8 dashboard visuals • Operational baseline

No financial savings or outcome improvement is established by this analysis.